

RESEARCH ARTICLE | MAY 15 1994

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J. Chem. Phys. 100, 7410–7415 (1994)

<https://doi.org/10.1063/1.466884>



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Benchmark calculations with correlated molecular wave functions.

IV. The classical barrier height of the $\text{H}+\text{H}_2\rightarrow\text{H}_2+\text{H}$ reaction

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(Received 26 July 1993; accepted 20 January 1994)

Using systematic sequences of correlation consistent Gaussian basis sets from double to sextuple zeta quality, the classical barrier height of the $\text{H}+\text{H}_2$ exchange reaction has been calculated by multireference configuration interaction (MRCI) methods. The MRCI calculations for collinear H_3 have also been calibrated against large basis set full CI (FCI) results, which demonstrate that the MRCI treatment leads to energies less than 1 $\mu\text{hartree}$ (<0.001 kcal/mol) above the FCI energies. The dependence of both the H_2 and H_3 total energies on the basis set is found to be very regular, and this behavior has been used to extrapolate to the complete basis set (CBS) limits. The resulting estimate of the H–H–H CBS limit yields a classical barrier height, relative to exact $\text{H}+\text{H}_2$, of 9.60 ± 0.02 kcal/mol; the best directly calculated value for the barrier is equal to 9.62 kcal/mol. These results are in excellent agreement with recent quantum Monte Carlo calculations.

I. INTRODUCTION

Modeling chemical systems by *ab initio* electronic structure techniques often requires a sophisticated treatment of electron correlation. The accuracy of the results, however, are still dependent on the quality of the one-particle basis set used to represent the molecular orbitals. A small basis set used with a highly correlated method can yield results of lesser accuracy than those obtained at lower levels of theory with larger basis sets. This interplay of errors between the correlation treatment and orbital basis can lead to erratic results which limits our understanding of the true accuracy of a given theoretical method. Only when the complete basis set (CBS) limit is approached can the *inherent* accuracy of a theoretical method be assessed.

In previous papers of this series we have investigated the inherent accuracy of various configuration interaction methods for the diatomic hydrides of the first¹ and second row² and the first row homonuclear diatomics.³ This work was made possible by using the family of correlation consistent Gaussian basis sets developed by Dunning and co-workers.^{4–6} These sets are compact, converge systematically to the CBS limit, and are well defined with respect to increases in both size and accuracy. In these basis sets, correlating *s*, *p*, *d*, ... functions are added in shells to the atomic Hartree–Fock orbitals with each function in a shell contributing a nearly equal amount to the correlation energy. Our previous studies^{1–3} have shown that by using systematically larger correlation consistent basis sets, most molecular properties converge to a well-defined asymptotic limit. The *intrinsic* error in a given property resulting from use of the chosen electronic structure method is obtained by comparing this estimated CBS limit to experimental results.

For molecular systems that are amenable to exact or

nearly exact electron correlation treatments, e.g., by the full configuration interaction (FCI) technique, a determination of the complete basis set limit should yield essentially the exact nonrelativistic value of the property of interest (within the Born–Oppenheimer approximation). These systems provide a sensitive test of the convergence of the correlation consistent basis sets to the CBS limit. One such system is the $\text{H}+\text{H}_2$ hydrogen exchange reaction, which has become a benchmark for both theoretical and experimental studies of chemical reactivity. For the reactants, nearly exact Born–Oppenheimer potential energy curves have been previously calculated for the ground state of the H_2 molecule by Kolos and Wolniewicz⁷ and more recently by Kolos, Szalewicz, and Monkhorst (KSM).⁸ For the H_3 species corresponding to the transition state in the $\text{H}+\text{H}_2\rightarrow\text{H}_2+\text{H}$ exchange reaction, several highly accurate studies have been carried out using both analytic variational methods^{9–21} and quantum Monte Carlo (QMC) techniques.^{22–24} Some of the *ab initio* studies involved calculations at a large number of nuclear configurations,^{10,14,16} which were subsequently used to determine analytical representations of the potential energy surfaces. The most accurate are the LSTH,²⁵ DMBE,¹⁴ and BKMP¹⁶ surfaces.

Previously, the lowest analytic variational energy for the collinear H–H–H transition state geometry was obtained by Liu¹¹ in large MRCI calculations with a (5*s*3*p*2*d*2*f*) Slater basis set. These calculations yielded a barrier height (at $r=1.759$ bohr) of 9.65 kcal/mol (9.86 kcal/mol relative to exact $\text{H}+\text{H}_2$), and an estimate of 9.59 ± 0.06 kcal/mol was derived for the true barrier by assuming the correlation error in H_3 was proportional to the correlation error in H_2 . Very recently, Cencek and Rychlewski²⁰ have carried out variational calculations on H_3 using explicitly correlated Gaussian functions. The energy they obtained with a 200 term expansion $-1.658\,965$ hartrees (at $r=1.75$ bohr), was lower than that of Liu's by 210 $\mu\text{hartree}$ (0.14 kcal/mol). The lowest energy reported for the $\text{H}+\text{H}_2$ transition state has been computed by Diedrich and Anderson²⁴ using the quantum Monte Carlo method. Their calculated total energy of

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^{b)}The Pacific Northwest Laboratory is operated for the U.S. Department of Energy by Battelle Memorial Institute under contract DE-AC06-76RLO 1830.

$-1.659\,154(14)$ hartrees at $r=1.757$ bohr yielded a barrier height relative to exact $\text{H}+\text{H}_2$ of 9.61 ± 0.01 kcal/mol. This result was in good agreement with the previous QMC result of 9.65 ± 0.08 kcal/mol obtained by Ceperley and Alder²² as well as the earlier estimate of Liu.¹¹

In the present study we have calculated the potential energy curves of the H_2 molecule and linear, symmetric H_3 using both full configuration interaction and large multireference CI (MRCI) wave functions. For each species a series of correlation consistent basis sets that varied from double to sextuple zeta quality was used. Correlation consistent basis sets that included extra diffuse functions in each angular momentum symmetry⁵ have also been investigated. The observed regularity of the basis set dependence leads to accurate extrapolations to the estimated CBS limits at the FCI (H_2) and MRCI (H_3) levels of theory. These results have been used to estimate the classical barrier height for the exchange reaction with an accuracy comparable to the recent QMC calculations of Diedrich and Anderson.²⁴

II. DETAILS OF THE CALCULATIONS

All of the present calculations on H , H_2 , and H_3 employed the MOLPRO92 suite of *ab initio* electronic structure programs.²⁶ Several correlation consistent Gaussian basis sets have been used in the present work. These sets ranged from a valence double-zeta, cc-pVDZ, to a valence sextuple-zeta set, cc-pV6Z. More specifically, they consisted of generally contracted²⁷ functions of the following sizes: cc-pVDZ-(4s1p)/[2s1p],⁴ cc-pVTZ-(5s2p1d)/[3s2p1d],⁴ cc-pVQZ-(6s3p2d1f)/[4s3p2d1f],⁴ cc-pV5Z-(8s4p2f1g)/[5s4p3d2f1g], and cc-pV6Z-(10s5p4d3f2g1h)/[6s5p4d3f2g1h]. Since diffuse functions can be important for molecular transition states, we have also utilized the same series as above with the addition of an extra diffuse function in each angular momentum symmetry. These functions have exponents optimized⁵ for the ground state of H^- and are designated by aug-cc-pVXZ, where X is the cardinal number of the basis set ($X=D, T, Q, 5, 6$). Due to the resulting size of the cc-pV6Z and aug-cc-pV6Z basis sets for H_3 (273 and 381 functions, respectively), the full sets could not be used. (The current MOLPRO integral program is limited to 255 basis functions.) For calculations on H_3 with the cc-pV6Z set, the h functions were deleted and the resulting set is designated cc-pV6Z(- h). It was simply not possible to carry out calculations on H_3 using the aug-cc-pV6Z set. For the H_2 molecule the only change from the full basis sets was the deletion of the diffuse f function from the aug-cc-pV6Z set due to numerical difficulties in the integral code; this set is designated aug-cc-pV6Z(- f). All of the basis sets used for H_3 were also employed in calculations on H_2 . The exponents, contraction coefficients, and atomic energies for the basis sets not previously published, i.e., cc-pV5Z, cc-pV6Z, aug-cc-pV5Z and aug-cc-pV6Z, are available upon request from the authors.²⁸

For the H_2 molecule, singles and doubles CI (full CI) calculations were carried out in D_{2h} symmetry with the above basis sets for seven geometries that covered the range $-0.3a_0 \leq r-r_{\text{exp}} \leq +0.5a_0$. Spectroscopic constants were

obtained by fitting the computed energies to sixth-order polynomials in $\Delta r=r-r_e$, followed by the usual Dunham analysis.²⁹ In the case of the aug-cc-pV6Z(- f) basis set, only the energy at $r=1.401$ bohr was computed due to difficulties of the type mentioned above.

For the linear H_3 system, both full configuration interaction and multireference CI have been carried out in the present work. For the FCI work, the determinant-based method of Knowles and Handy^{30,31} was used. Two different reference functions, both based on complete active space self-consistent-field (CASSCF) orbitals,^{32,33} were used in the MRCI calculations.^{34,35} One consisted of a CAS expansion among the three molecular orbitals formed from the hydrogen $1s$ atomic orbitals. This full valence CAS consisted of just four configuration state functions (CSFs); this is the GVB wave function for H_3 . The second function was designed to recover as much of the FCI energy as possible. Test calculations with the aug-cc-pVQZ basis set showed that MRCI wave functions based on a CAS expansion that included $6\sigma_g$, $4\sigma_u$, $2\pi_u$, and $1\pi_g$ orbitals (in D_{2h} symmetry: $1-6a_g$, $1-2b_{2u}$, $1-2b_{3u}$, $1-4b_{1u}$, $1b_{2g}$, $1b_{3g}$), which yielded 252 CSFs, reproduced the FCI energy to within $0.2\,\mu\text{hartree}$ (0.0001 kcal/mol). A similar MRCI approach was used previously by Bauschlicher, Langhoff, and Partridge¹⁵ and more recently by Partridge *et al.*²¹ The FCI and MRCI calculations for H_3 were carried out in D_{2h} symmetry at seven geometries over the range $-0.3a_0 \leq r_{\text{HH}}-1.757a_0 \leq +0.5a_0$. For H_3 , the polynomial fits in the symmetry coordinate $(r_1+r_2)/\sqrt{2}$ and subsequent calculations of the transition state geometry and stretching harmonic frequencies were carried out with the program SURFIT.³⁶

By systematically increasing the size of the one-particle basis set within the family of correlation consistent sets, regular convergence toward the complete basis set limit has been observed for quantities such as the minimum energy, equilibrium bond distances, and harmonic frequencies.¹⁻³ Previously, exponential functions of the form

$$A(X)=A(\infty)+Be^{-CX}, \quad (1)$$

where X is the cardinal number of the basis set (2, 3, 4, 5 for DZ, TZ, QZ, and 5Z sets, respectively) and $A(\infty)$ is the estimated CBS limit as $X \rightarrow \infty$, have been successfully used to represent the basis set dependence of E_e , r_e , and ω_e for selected diatomic molecules involving both first and second row atoms.¹⁻³ In the present case, however, we have found that the function

$$A(X)=A(\infty)+Be^{-(X-1)}+Ce^{-(X-1)^2} \quad (2)$$

provides a much better representation of the calculated energies. Using Eq. (2), the estimated CBS limits for E_e have been estimated for both H_2 and H_3 for each method. By comparing the calculated minimum H_3 energies to the "exact" Born-Oppenheimer energies of H_2 (Ref. 8) and the hydrogen atom (-0.500 hartrees), the estimated CBS limit (at essentially the FCI level) for the classical barrier height of the $\text{H}+\text{H}_2$ reaction has been derived.

Finally, it should be noted that the classical barrier height obtained in the above manner does not include a cor-

TABLE I. Full CI results for separated H+H₂ compared to the exact Born–Oppenheimer potential for H₂ with a hydrogen atom infinitely far away.

Method	Basis set	$E(H)$ (hartrees)	$E_e(H_2)$ (hartrees)	$E_e(H_2+E(H))$ (hartrees)	D_e (kcal/mol)	r_e (Å)	ω_e (cm ⁻¹)	$\omega_e x_e$ (cm ⁻¹)
Exact ^a		-0.500 000	-1.174 476	-1.674 476	109.485	0.74143	4404.70	121.72
FCI	cc-pVDZ	-0.499 278	-1.163 673	-1.662 951	103.61	0.7609	4383.3	117.8
	cc-pVTZ	-0.499 810	-1.172 337	-1.672 147	108.38	0.7426	4409.2	120.9
	cc-pVQZ	-0.499 946	-1.173 796	-1.673 742	109.13	0.7419	4403.2	119.0
	cc-pV5Z	-0.499 995	-1.174 222	-1.674 217	109.33	0.7415	4404.5	119.5
	cc-pV6Z(-h)	-0.499 999	-1.174 318	-1.674 317	109.39	0.7415	4404.6	119.9
	cc-pV6Z	-0.499 999	-1.174 342	-1.674 341	109.40	0.7415	4404.5	119.8
	est. CBS limit (2-6) ^b		-1.174 454(47)	-1.674 469(61)	109.45			
	est. CBS limit (3-6) ^c		-1.174 440(61)	-1.674 452(87)	109.46			
	aug-cc-pVDZ	-0.499 334	-1.164 900	-1.664 234	104.31	0.7617	4345.3	117.3
	aug-cc-pVTZ	-0.499 821	-1.172 636	-1.672 457	108.56	0.7430	4400.9	123.4
	aug-cc-pVQZ	-0.499 948	-1.173 867	-1.673 815	109.17	0.7420	4399.7	119.4
	aug-cc-pV5Z	-0.499 995	-1.174 252	-1.674 247	109.35	0.7416	4403.1	119.6
	aug-cc-pV6Z(-f) ^d	-0.499 999	-1.174 354	-1.674 353	109.41			
	est. CBS limit (2-5) ^e		-1.174 454(48)	-1.674 474(53)	109.45			
	est. CBS limit (3-5) ^f		-1.174 475	-1.674 497	109.45			

^aThe minimum H₂ energy was derived from a polynomial fit to seven points of the Kolos, Szalewicz, and Monkhorst H₂ potential (Ref. 8) which spanned a similar range of bond distances as in our present calculations. The exact Born–Oppenheimer energy for the hydrogen atom is taken to be -0.500 000 hartrees.

^bObtained by fitting the cc-pVDZ through cc-pV6Z values to Eq. (2). See the text.

^cObtained by fitting the cc-pVTZ through cc-pV6Z values to Eq. (2). See the text.

^dCalculated only for $r=1.401$ bohr.

^eObtained by fitting the aug-cc-pVDZ through aug-cc-pV5Z values to Eq. (2). See the text.

^fObtained by fitting the aug-cc-pVTZ through aug-cc-pV5Z values to Eq. (2); for these extrapolations the fit is exact.

rection for the finite mass of the hydrogen nucleus (see, e.g., Ref. 37). This error (~ 60 cm⁻¹ for the hydrogen atom) actually exceeds the errors associated with the electronic energies obtained from the current calculations (~ 7 cm⁻¹) as well as those of Diedrich and Anderson²⁴ (~ 3 cm⁻¹). This effect, along with adiabatic, nonadiabatic, relativistic, and radiative corrections, must, of course, be included when comparing to experiment but have not been included here to facilitate comparison with the previous calculations.

III. RESULTS AND DISCUSSION

A. Calculations on H₂+H

Table I compares our FCI results for H₂ and hydrogen atom with the “exact” potential calculated by Kolos, Szalewicz, and Monkhorst.⁸ Excellent agreement with the KSM potential is observed with the cc-pVQZ and aug-cc-pVQZ basis sets and the spectroscopic constants calculated with the V5Z sets are nearly identical to the KSM values. While the V6Z sets do not result in significantly better spectroscopic constants, the total energies continue their steady progression towards the exact Born–Oppenheimer value. With the aug-cc-pV6Z(-f) basis set the computed FCI energy (H₂+H) of -1.674 353 hartrees is just 123 μ hartrees (0.08 kcal/mol) above the exact nonrelativistic value. Figure 1 shows the convergence of our calculated H+H₂ energies toward the exact KSM value with increasing basis set quality. Exponential behavior is evident, and the fits to Eq. (2), which are also shown in Fig. 1, yield CBS limits for E_e that differ from the exact value by just 7 ± 60 μ hartrees (0.004 ± 0.04 kcal/mol) and 2 ± 50 μ hartrees (0.001 ± 0.03 kcal/mol) for the cc-pVXZ

and aug-cc-pVXZ basis set series, respectively. In both cases the quoted uncertainty is three times the error estimated from the nonlinear least squares fits.

B. Calculations on the H–H–H saddle point

Our FCI and MRCI results for the collinear H₃ symmetric stretch potential are listed in Table II, together with se-

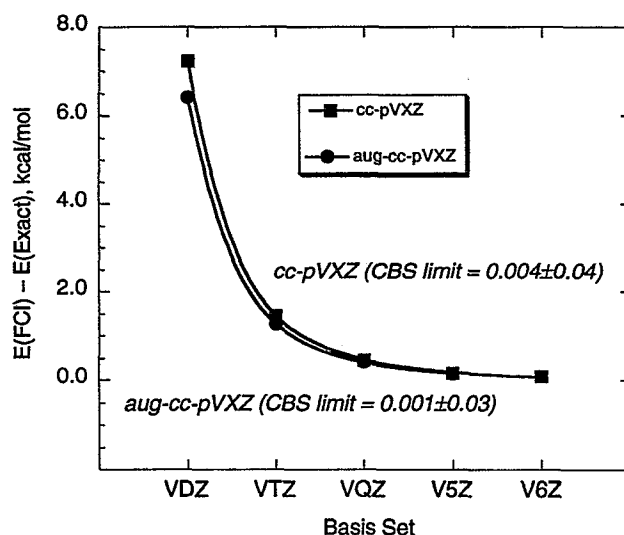


FIG. 1. The convergence of the separated H+H₂ FCI energy as a function of the correlation consistent basis set used is shown relative to the exact energy ($-1.674 476$ hartrees⁸). The solid lines are fits to Eq. (2).

TABLE II. FCI and MRCI results for the collinear H+H₂ barrier compared to selected previous analytic variational and quantum Monte Carlo results.

Method	Basis set	$E_e(\text{H-H-H})$ (hartrees)	E_{b1}^a (kcal/mol)	E_{b2}^b (kcal/mol)	r_e (bohr)	ω_{sym} (cm ⁻¹)
MRCI ^c	5s3p2d2f(Slater)	-1.658 755	9.65	9.865	1.759	
MRCI ^d	6s6p4d2f	-1.658 74	9.649			
MRCI ^d	6s6p4d3f1g		9.632			
HCI ^e	200 terms	-1.658 965		9.733		
QMC ^f		-1.659 157(14)	9.60	9.61(1)	1.757	2040
FCI	cc-pVDZ	-1.646 6211	10.247	17.479	1.7826	2074.3
	cc-pVTZ	-1.656 3085	9.939	11.400	1.7575	2053.3
	cc-pVQZ	-1.658 2623	9.713	10.174	1.7575	2053.6
	cc-pV5Z ^g	-1.658 8296	9.656	9.818		
	aug-cc-pVDZ	-1.649 4646	9.268	15.695	1.7859	2043.8
	aug-cc-pVTZ	-1.657 0142	9.691	10.957	1.7600	2043.3
	aug-cc-pVQZ	-1.658 4703	9.629	10.044	1.7582	2050.0
MRCI-4	cc-pVDZ	-1.646 5154	10.314	17.546	1.7829	2073.4
	cc-pVTZ	-1.656 1221	10.056	11.517	1.7580	2052.2
	cc-pVQZ	-1.658 0477	9.848	10.309	1.7579	2052.3
	cc-pV5Z	-1.658 6064	9.796	9.958	1.7576	2051.4
	cc-pV6Z(-h)	-1.658 7383	9.776	9.876	1.7576	2050.4
	est. CBS limit ^h	-1.658 971(92)		9.73(6)		
	aug-cc-pVDZ	-1.649 3072	9.367	15.794	1.7861	2042.9
	aug-cc-pVTZ	-1.656 7996	9.825	11.092	1.7604	2042.2
	aug-cc-pVQZ	-1.658 2459	9.770	10.185	1.7586	2048.8
	aug-cc-pV5Z	-1.658 6819	9.767	9.911	1.7579	2049.9
	est. CBS limit ⁱ	-1.658 948(31)		9.74(2)		
	cc-pVDZ	-1.646 6211	10.247	17.479	1.7826	2074.3
	cc-pVTZ	-1.656 3085	9.939	11.400	1.7575	2053.3
MRCI-252	cc-pVQZ	-1.658 2622	9.713	10.174	1.7575	2053.6
	cc-pV5Z	-1.658 8294	9.656	9.818	1.7572	2052.6
	cc-pV6Z(-h)	-1.658 9596	9.637	9.737	1.7572	2051.6
	est. CBS limit (2-5) ^h	-1.659 200(95)		9.59(6)		
	est. CBS limit (3-5) ^j	-1.659 158		9.61		
	aug-cc-pVDZ	-1.649 4646	9.268	15.695	1.7859	2043.8
	aug-cc-pVTZ	-1.657 0141	9.691	10.957	1.7600	2043.3
	aug-cc-pVQZ	-1.658 4701	9.629	10.044	1.7582	2050.0
	aug-cc-pV5Z	-1.658 9082	9.625	9.769	1.7575	2051.1
	est. CBS limit (2-5) ⁱ	-1.659 176(32)		9.60(2)		
	est. CBS limit (3-5) ^k	-1.659 162		9.61		

^aThe H+H₂ collinear barrier height computed relative to the same calculation on separated H+H₂.^bThe H+H₂ collinear barrier height computed relative to the exact energy of -1.674 476 hartrees for separated H+H₂ ($r=1.401\ 09$ bohr).^cReference 11.^dReference 21.^eHylleraas-CI calculation at $r=1.75$ bohr using explicitly correlated Gaussians (Ref. 20).^fReference 24. The symmetric harmonic frequency shown here was derived by a weighted least-squares fit to the computed energies of Ref. 24.^gComputed only at $r=1.7571$ bohr.^hObtained by fitting the cc-pVDZ through cc-pV5Z values to Eq. (2). The uncertainties in the last digit are three times the estimated error from the least-squares fit.ⁱObtained by fitting the aug-cc-pVDZ through aug-cc-pV5Z values to Eq. (2). The uncertainties in the last digit are three times the estimated error from the least-squares fit.^jObtained by fitting the cc-pVTZ through cc-pV5Z values to Eq. (2).^kObtained by fitting the aug-cc-pVTZ through aug-cc-pV5Z values to Eq. (2).

lected previous QMC and analytic variational results. The saddle point energy from the MRCI-252 calculations with the cc-pV6Z(-h) basis set, -1.658 960 hartrees, is 204 μ hartrees (0.12 kcal/mol) below the energy reported by Liu,¹¹

only 5 μ hartrees (0.003 kcal/mol) higher than the variational energy obtained using explicitly correlated Gaussian functions,²⁰ and 197 μ hartrees (0.12 kcal/mol) higher than the QMC value of Diedrich and Anderson.²⁴ The *extrapo-*

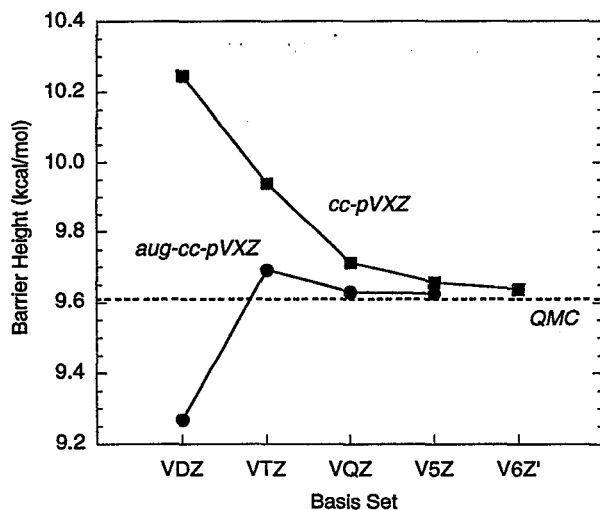


FIG. 2. The $\text{H}+\text{H}_2$ MRCI-252 barrier height as a function of basis set is shown relative to FCI calculations on separated $\text{H}+\text{H}_2$ with the identical basis. The line at 9.61 kcal/mol is from the QMC calculation of Diedrich and Anderson (Ref. 24).

lated energies for the cc-pVXZ and aug-cc-pVXZ sets are 43 and 19 $\mu\text{hartrees}$, or 0.03 and 0.01 kcal/mol, respectively, below the QMC energy. In good agreement with previous work, the D_{2h} transition state has a calculated bond length equal to 1.7575 bohr at the CMRCI-252/aug-c-pV5Z level. The same calculations also result in a symmetric stretching frequency of 2051.1 cm^{-1} at the saddle point. To ascertain the imaginary frequency of the asymmetric stretching coordinate, additional calculations were carried out with the aug-cc-pVDZ, aug-cc-pVTZ, and aug-cc-pVQZ basis sets in C_{2v} symmetry (510 reference CSFs in the MRCI). Fits to grids of 11 energies for each basis set yielded asymmetric stretch harmonic frequencies of 1442.4i, 1503.4i, and 1498.4i cm^{-1} , respectively.

Figure 2 depicts the MRCI-252 barrier heights computed relative to the FCI calculations with the same basis sets on separated $\text{H}+\text{H}_2$ (Table I). Smooth convergence is observed, which indicates that the energies for the reactant and saddle point species are converging at nearly the same rate. The one exception is the result for the aug-cc-pVDZ basis set; for this set there is a distinct bias toward the H_3 saddle point, resulting in a barrier height much smaller than the values computed with the more accurate basis sets. Neglecting the calculations with the aug-cc-pVDZ set, the augmented sets exhibit a faster convergence to the basis set limit than the standard sets. As can be seen in Table II, the barrier height from the MRCI-252 calculations with the aug-cc-pV5Z basis set, 9.625 kcal/mol, is lower than the cc-pV5Z value by 0.03 kcal/mol. Furthermore, the difference in the barrier heights calculated with the aug-cc-pVQZ and aug-cc-pV5Z sets (relative to exact $\text{H}+\text{H}_2$) is just 0.275 kcal/mol, compared to a difference of 0.356 kcal/mol with the standard sets.

Figure 3 is a plot of the convergence of the $\text{H}+\text{H}_2$ barrier height relative to the exact separated $\text{H}+\text{H}_2$ energy of $-1.674\,476$ hartrees.⁸ This figure is equivalent to plotting

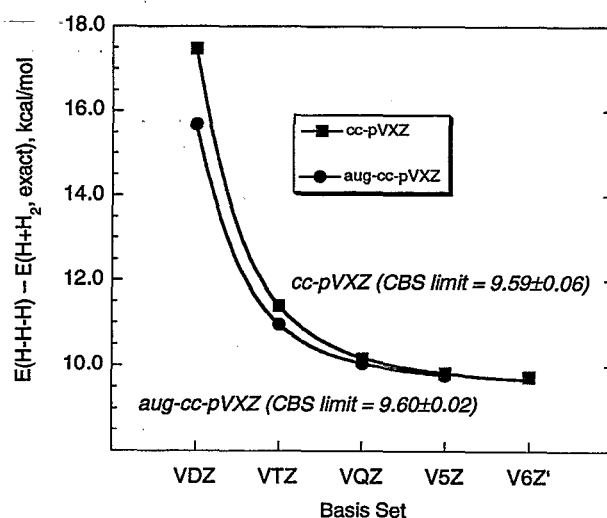


FIG. 3. The convergence of the $\text{H}+\text{H}_2$ MRCI-252 barrier height relative to exact $\text{H}+\text{H}_2$ is shown as a function of the correlation consistent basis set used. The solid lines are fits to Eq. (2). The cc-pV6Z(-h) value was not included in the fit.

the H_3 total energies and hence is analogous to Fig. 1. Figure 3 demonstrates smooth exponential convergence of the MRCI-252 barrier height for both the standard and augmented correlation consistent basis sets. The fits to Eq. (2), which are also shown in the figure, yield CBS limits of 9.59 ± 0.06 kcal/mol (cc-pVXZ, $X=D, T, Q, 5$) and 9.60 ± 0.02 kcal/mol (aug-cc-pVXZ, $X=D, T, Q, 5$), where the quoted uncertainties are, again, three times the estimated errors of the fits. The fits based on just the TZ, QZ, and 5Z energies yield barrier heights of 9.61 kcal/mol, in excellent agreement with the values obtained using all four basis sets. Our best estimate for the barrier height relative to exact $\text{H}+\text{H}_2$, 9.60 ± 0.02 kcal/mol, is in very good agreement with the QMC value of 9.61 ± 0.01 kcal/mol.²⁴

Using our exponential fit to the aug-cc-pVXZ barrier heights, an estimate of the basis set required to reach a specified error limit can be derived. Extrapolation of the fit by Eq. (2) yields predicted barrier heights (relative to exact $\text{H}+\text{H}_2$) of 9.66 kcal/mol (aug-cc-pV6Z, 381 contracted functions), 9.62 kcal/mol (aug-cc-pV7Z, 567 functions) and 9.61 kcal/mol (aug-cc-pV8Z, 804 functions). Thus, an aug-cc-pV7Z basis set would be required to yield a total energy for the $\text{H}-\text{H}-\text{H}$ barrier within 0.02 kcal/mol of the exact Born-Oppenheimer value. To estimate the time required to carry out this calculation, the observed timings for the aug-cc-pVXZ ($X=D, T, Q, 5$) basis set calculations were employed. These scaled approximately as N^4 (Note that this is certainly not true for all full CI problems), where N is the number of basis functions. Extrapolating our timings for an IBM RS/6000 550 workstation yields an estimated CPU requirement of just over 90 h for a MRCI-252 calculation with an aug-cc-pV7Z basis set (18 h for the aug-cc-pV6Z basis and just 364 h for an aug-cc-pV8Z basis). This is much smaller than the 10–20 years as recently proposed by Clem-

enti *et al.*¹⁷ for a Hylleraas-CI calculation of similar accuracy.

Finally, we note that the differences between the valence GVB-CI (MRCI-4) and the extended reference MRCI (MRCI-252) are surprisingly small. The barrier heights computed with the MRCI-4 wave functions are just 0.10–0.14 kcal/mol higher than those computed with FCI or MRCI-252, and the curves are nearly parallel.

IV. CONCLUSIONS

Accurate configuration interaction calculations have been carried out on molecular H₂ and the collinear H₃ transition state of the H+H₂ exchange reaction. Due to the smooth convergence behavior observed with the correlation consistent basis sets toward the complete basis set limit, extrapolations to the exact Born–Oppenheimer total energies for H₂ and H₃ were possible. The extrapolated H₃ energy has been combined with the known (exact) energy of separated H+H₂ to yield an accurate estimate of the classical barrier height for the exchange reaction of 9.60±0.02 kcal/mol. This is in excellent agreement with the results of recent quantum Monte Carlo calculations.

ACKNOWLEDGMENTS

This work was supported by the Division of Chemical Sciences in the Office of Basic Energy Sciences of the U.S. Department of Energy at Pacific Northwest Laboratory under Contract No. DE-AC06-76RLO 1830. K.A.P. and D.E.W. acknowledge the support of the Northwest College and University Association for Science (Washington State University) under Grant No. DE-FG06-89ER-75522 with the U.S. Department of Energy. Computational resources for this work were provided by the Division of Chemical Sciences and by the Office of Scientific Computing, Office of Energy Research, at the National Energy Research Supercomputing Center (Livermore, California).

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